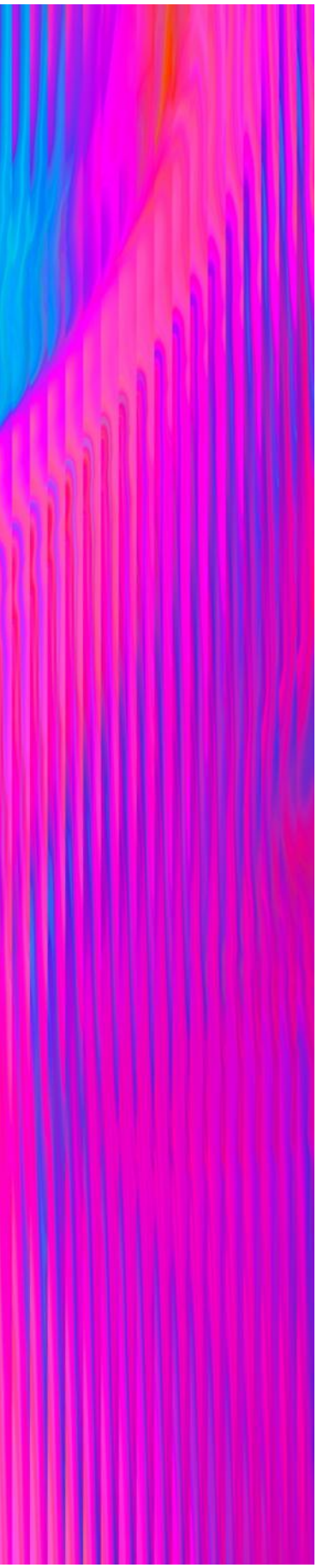


NAUKRI JOB TRENDS PROJECT REPORT

OCTOBER 2026

BY DHARSHAN ROY K
[ANP-D 7635]

1. Executive summary

This project analyses 23,201 Naukri.com postings for six data and analytics role families across 11 Indian city searches. A three-page Power BI dashboard presents demand by role, experience, city and work mode, together with salary benchmarks, employer patterns and a state map. A separate cleaning exercise produced a de-duplicated star schema of 21,123 postings.

Headline findings (all figures computed from the raw CSV):

- **Demand is led by data science and analytics roles.** Data Scientist (6,455), Data Analyst (4,729) and Business Analyst (4,505) account for 67.6% of postings.
- **Mid-level talent is most wanted.** Mid (3-5 years) roles are 43.6% of postings; Fresher-tier roles are 9.6%.
- **Salary transparency is low.** Only 11.9% of postings (2,768) disclose a salary. The average disclosed midpoint is 15.32 LPA.
- **Geography is concentrated.** Bangalore, Mumbai and Chennai searches account for 42.0% of postings, but scrape coverage per city is uneven.
- **Most roles are on-site.** On-site 80.6%, Hybrid 10.1%, Remote 9.3%.
- **Python is the most requested skill** (4,266 mentions), followed by data analysis, machine learning and SQL.

Key cautions

- The dashboard reads the **raw** table (23,201 rows, 6,993 employer names), not the cleaned 21,123-row model, so it includes 2,078 duplicate postings.
- 306 postings marked "Unpaid" carry a 10-18 LPA salary and are counted as disclosed, which affects salary averages.
- The Page 3 takeaway says Fresher openings are 18.0%; that is the fresher-friendly flag, whereas the Fresher tier is 9.6%.

2. Background and objectives

Hiring in India's data and analytics sector is fragmented across portals, and job adverts rarely publish pay. The project aims to give two audiences a single, defensible view of the market: university faculty, who need documented methodology and data governance, and industry leaders, who need clear indicators of demand, entry-level opportunity and compensation.

- Quantify demand by role family, experience level, city and work mode.
- Benchmark disclosed salaries and measure how transparent employers are.
- Document data quality so every figure can be traced and challenged.
- Deliver an interactive, filterable dashboard with an executive summary page.

3. Data sources

File	Content	Rows x cols
indian_tech_jobs_2026.csv	Raw scraped listings with 32 columns: 19 source fields and 13 derived fields (tiers, flags, midpoints).	23,201 x 32
Indian_Tech_Jobs_Cleaned.csv	Notes describing the cleaned Power BI star schema and cleaning rules (not row-level data).	14 lines
naukri_cleaned_dataset_2026.csv	Four headline KPIs: postings evaluated, unique employers, average salary, top skill.	4 x 2
Naukri_Job_trend_analysis_project _dashboard.pbix	Power BI report: 3 pages, 3 bookmarks, tables indian_tech_jobs_2026, state_name, Metric Selector.	-

The full column-level description is in the companion workbook **Naukri_Data_Dictionary.xlsx**.

4. Data preparation

The cleaning notes describe the following rules, which build the 21,123-row fact table:

Area	Rule
Duplicates	Rows sharing a job URL are removed (same posting scraped under several cities); lowest job_id kept; rows without URL kept. Check: 23,201 rows minus 2,078 repeat URLs = 21,123.
Salary	Re-derived from salary_raw. Undisclosed, Unpaid, open lower bound, invalid and outlier values become blank with a SalaryStatus. Monthly pay x12.
Experience	Re-derived from experience_raw; non-year text gets no range; bands use minimum years.
Recency	PostedDate only where an exact relative age exists; "3+ weeks ago" is a capped bucket.
City	Locality suffix removed; Bengaluru becomes Bangalore; region grouping defined by analyst.
Skills	Split, lower-cased, de-duplicated per posting; tags over 60 characters dropped.
Dropped columns	Constants (data_source, scraped_at, salary_negotiable), unreliable derived fields, multi-valued location, description and URL.

5. Data model

The documented star schema has a central fact table (one row per de-duplicated posting) and dimensions for employer, location, role, work mode, experience band, recency and date, plus a skill dimension linked through a many-to-many bridge table. A DQ ledger of live COUNTIF checks sits beside the fact table.

Table	Type	Grain
Fact_JobPosting	Fact	One row per de-duplicated posting (21,123)
Dim_Employer, Dim_Location, Dim_Role, Dim_WorkMode, Dim_Experience, Dim_Recency, Dim_Date	Dimensions	One row per member
Dim_Skill	Dimension	One row per distinct skill
Bridge_JobSkill	Bridge	One row per posting-skill pair
DQ_Ledger	Audit	Data-quality counts

The Power BI file itself loads a different model: the raw table indian_tech_jobs_2026 (all 32 columns), a table called state_name that holds the five measures and supplies state names for the map, and a Metric Selector parameter table. Its compressed data model could not be opened, so measure definitions are described by behaviour, not by DAX text.

Measures used in the dashboard

Measure	Meaning	Check against raw data
Total Jobs	Count of postings in the current filter	23,201 overall
Avg Benchmark Salary (LPA)	Average salary of postings that disclose pay	15.32 = mean midpoint of 2,768 disclosed rows
Disclosed Salary Rate	Share of postings that disclose pay	11.9%
Fresher Friendly Ratio	Share of postings flagged fresher friendly	18.0%
Avg Disclosed Salary	Average disclosed salary per employer (scatter and table)	-

6. Dashboard walkthrough

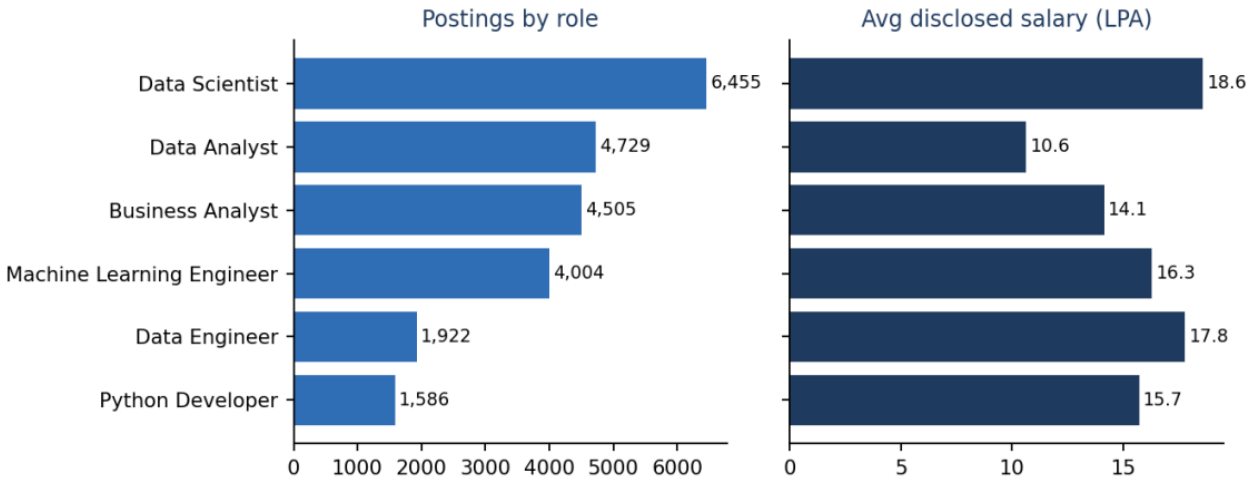
Page	Purpose	Main visuals
Page 1 - Overview	Executive view of demand and pay	KPI cards (Total Jobs, Avg Benchmark Salary, Disclosed Salary Rate, Fresher Friendly Ratio); experience funnel; work-mode pie; jobs by city; jobs by role (switchable with Metric Selector); salary by years of experience; pivot of experience tier by role.
Page 2 - Employers	Who is hiring and what they pay	Scatter of company rating vs postings (bubble = size); employer table with salary range; experience tier by skill domain.
Page 3 - Geography and insights	Where demand sits and key takeaways	India state shape map; state and experience combo; city and work-mode combo; size by salary tier; skills-count line; Executive Key Takeaways text box.

Interaction design

- Slicers: skill domain, salary tier, company size, work mode, salary range (LPA), role, city, experience tier.
- Metric Selector slicer switches the donut and role bar between metrics.
- Navigation buttons (home, info) and a Clear All Slicers button on each page.
- Bookmarks: Clear Filter, Insights, hide insights.

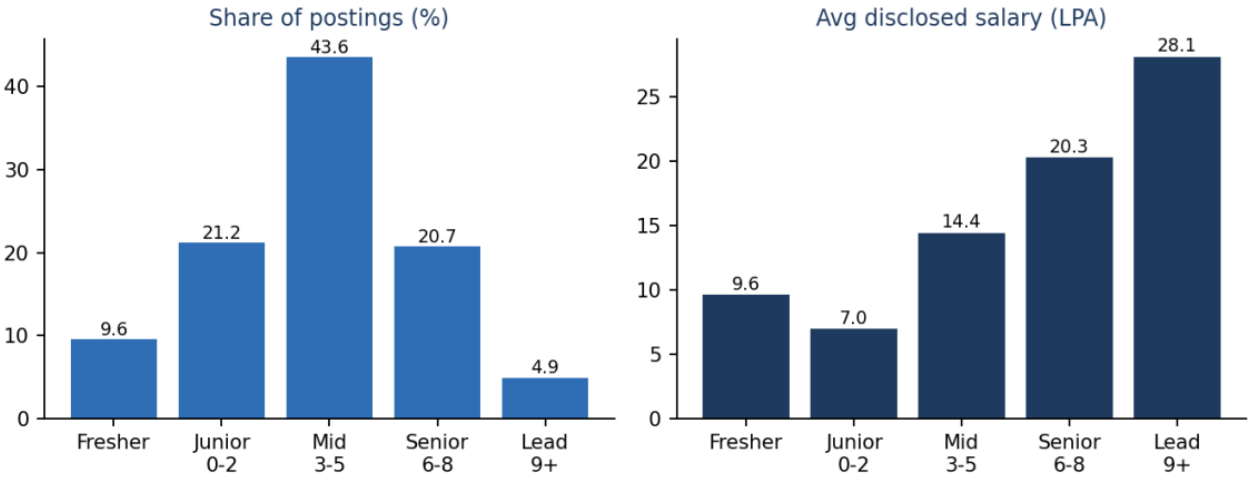
7. Key findings

7.1 Demand and pay by role



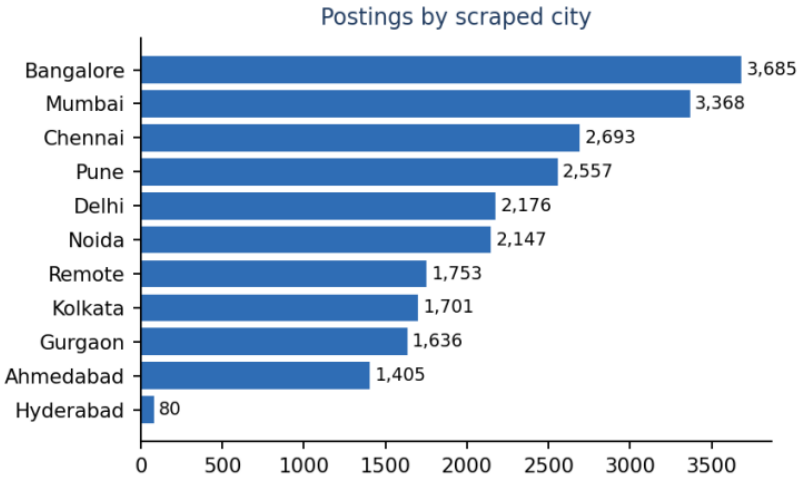
Data Scientist roles are both the most numerous and best paid among disclosed postings (18.6 LPA), followed by Data Engineer (17.9). Data Analyst roles are the lowest paid (10.6 LPA). Salary figures use only the 2,768 disclosed postings and are therefore indicative.

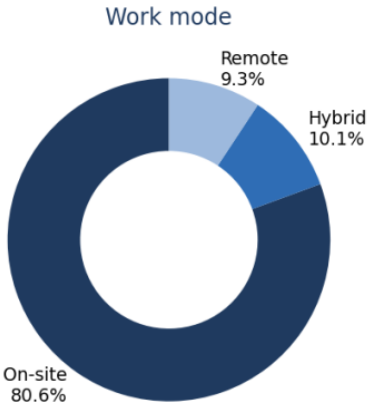
7.2 Experience profile



Mid-level roles dominate. Salary rises steadily with experience, from 9.6 LPA at Fresher to 28.3 LPA for Lead/Architect. The 306 unpaid rows sit mostly in the Mid tier (269); excluding them moves the Mid average only from 14.5 to 14.6 LPA, but the effect on the overall 15.32 LPA is larger (15.49 without them).

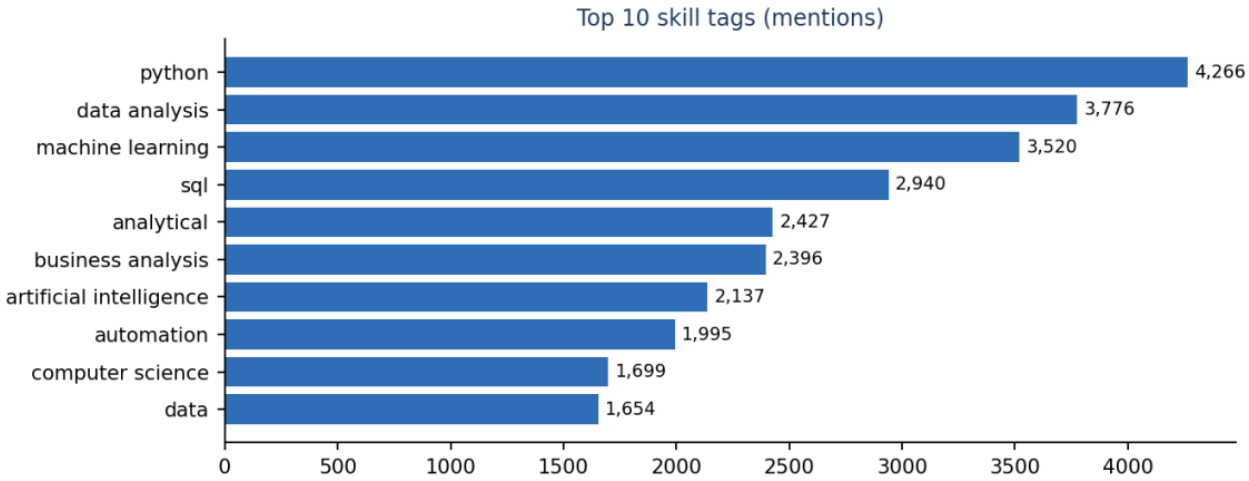
7.3 Location and work mode





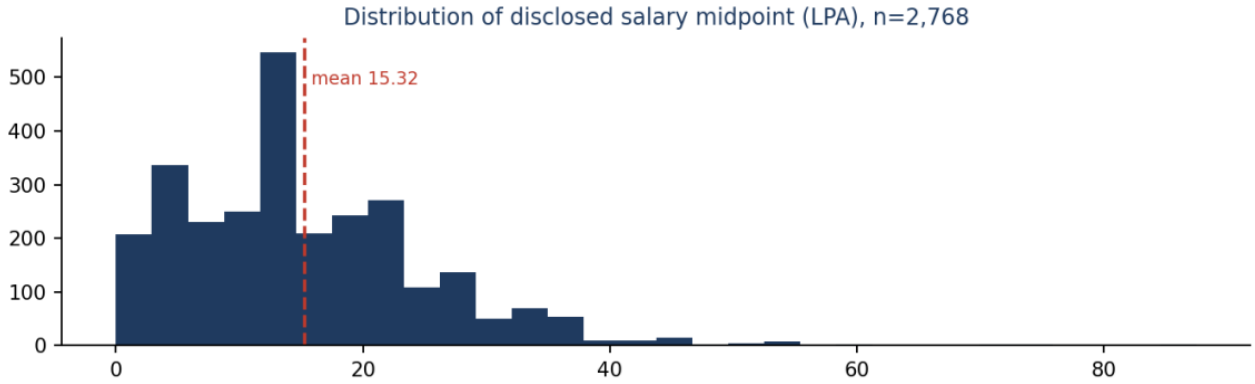
The top three searched cities hold 42.0% of postings. Hyderabad appears with only 80 rows, so city counts reflect scrape coverage as much as true demand. About 28% of postings list several cities.

7.4 Skills



Python leads, with data analysis, machine learning and SQL next. Skill tags are capped at 8 per posting (89.8% of rows reach the cap), so counts understate the long tail.

7.5 Salary distribution



8. Data quality and limitations

Issue	Evidence	Effect
Duplicate postings	2,078 repeat job URLs	Dashboard counts are 9.8% above the de-duplicated 21,123
"Unpaid" rows carry salary	306 rows: min 10, max 18, midpoint 14 LPA, flagged disclosed	Disclosed count and averages affected (15.49 LPA without them)
Undisclosed salary stored as 0	20,433 rows; averaging all rows gives 1.83 LPA	Misleading averages if not filtered
Date-corrupted experience	82 values such as "13-Jun"; 90 rows have min > max	Wrong experience tier for these rows
Placeholders as text	"Not Available": skills 547, description 404, posted date 49	Should be NULL
Capped recency	49% of rows are "3+ weeks ago"	No trend over time can be derived
Skewed company size	Small/Startup 81.3%; Mid (100-999) 0.3%	Size comparisons weak
Multi-valued location	27.8% list several cities; 198 primary_city values	City analysis uses scraped_city
Unclear flags	is_senior true for 26% of Freshers	Rule undocumented
Single scrape date	All rows 10 June 2025	Cross-sectional snapshot only; no time series

Methodological limits: the data is a sample of one portal on one day; postings are not hires; disclosed salaries come from a self-selecting 11.9% of employers and may not represent the rest.

Appendix A - Files delivered

File	Description
Naukri_Job_Trend_Analysis_Project_Report.pdf	This report
Naukri_Data_Dictionary.xlsx	Data dictionary: raw dataset, cleaned star schema, KPI file, Power BI model, lookups, quality log, lineage

Appendix B - Method note

All numbers in this report were computed directly from the uploaded CSV with Python (pandas). Report structure, page layout and fields were read from the .pbix report definition. DAX expressions could not be read because the data model is compressed; measure meanings were confirmed by reproducing their headline values from the raw data.